

Spatiotemporal coherence of Meridional Heat Transport in the Atlantic

Drivers of variability

Lea Sophie Dietz, 6th July 2026

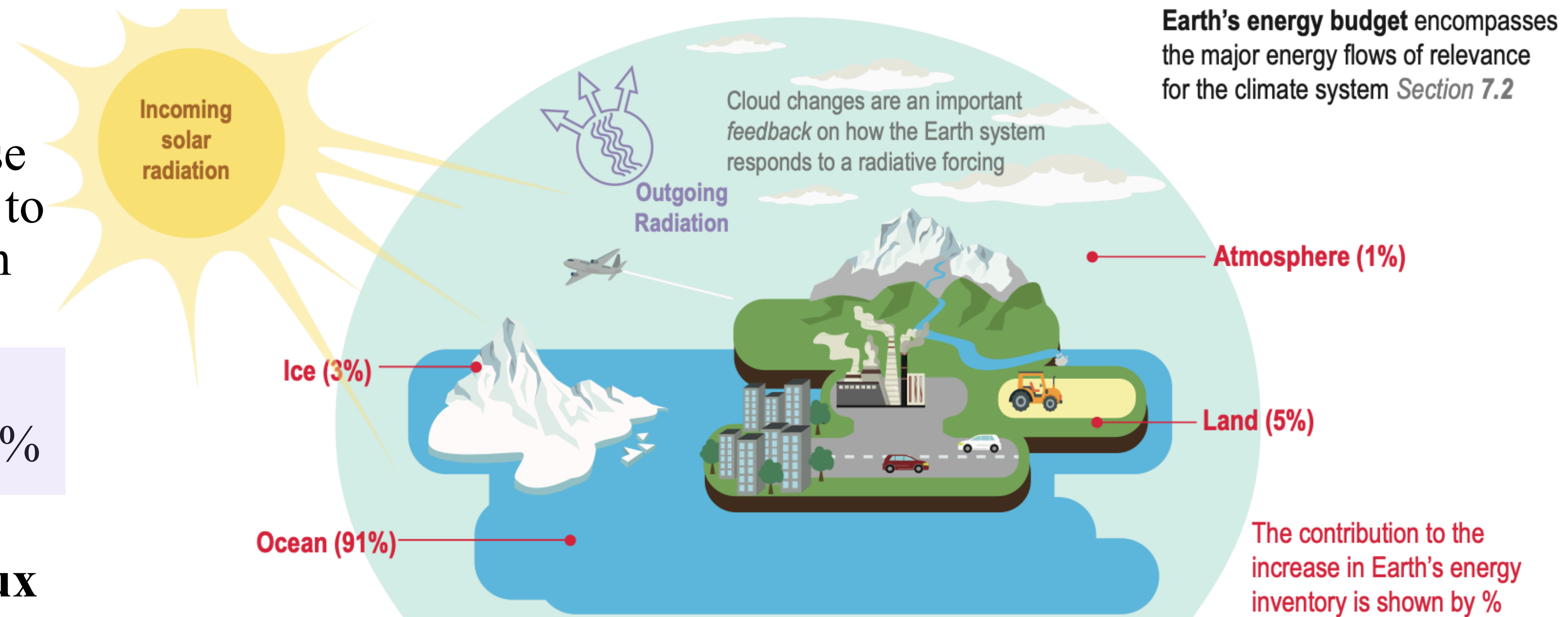
Supervisors: Eleanor Frajka-Williams, Lara Heyl

Motivation

Earth's energy budget in imbalance: What happens with heat stored in the ocean?

Earth Energy Imbalance (EEI):

- Anthropogenic greenhouse gas increases cause Earth to **absorb more energy** than it emits
- **Ocean is dominant heat sink** with an uptake of 91%
→ Excess heat enters the ocean via **surface heat flux** (F_s)



IPCC, 2021, WG I, Ch. 7

But what happens to this heat once it's in the ocean?

Motivation

From surface heat flux to ocean heat transport

- This heat is either **stored** or **recirculated**

(F_s here upwards)

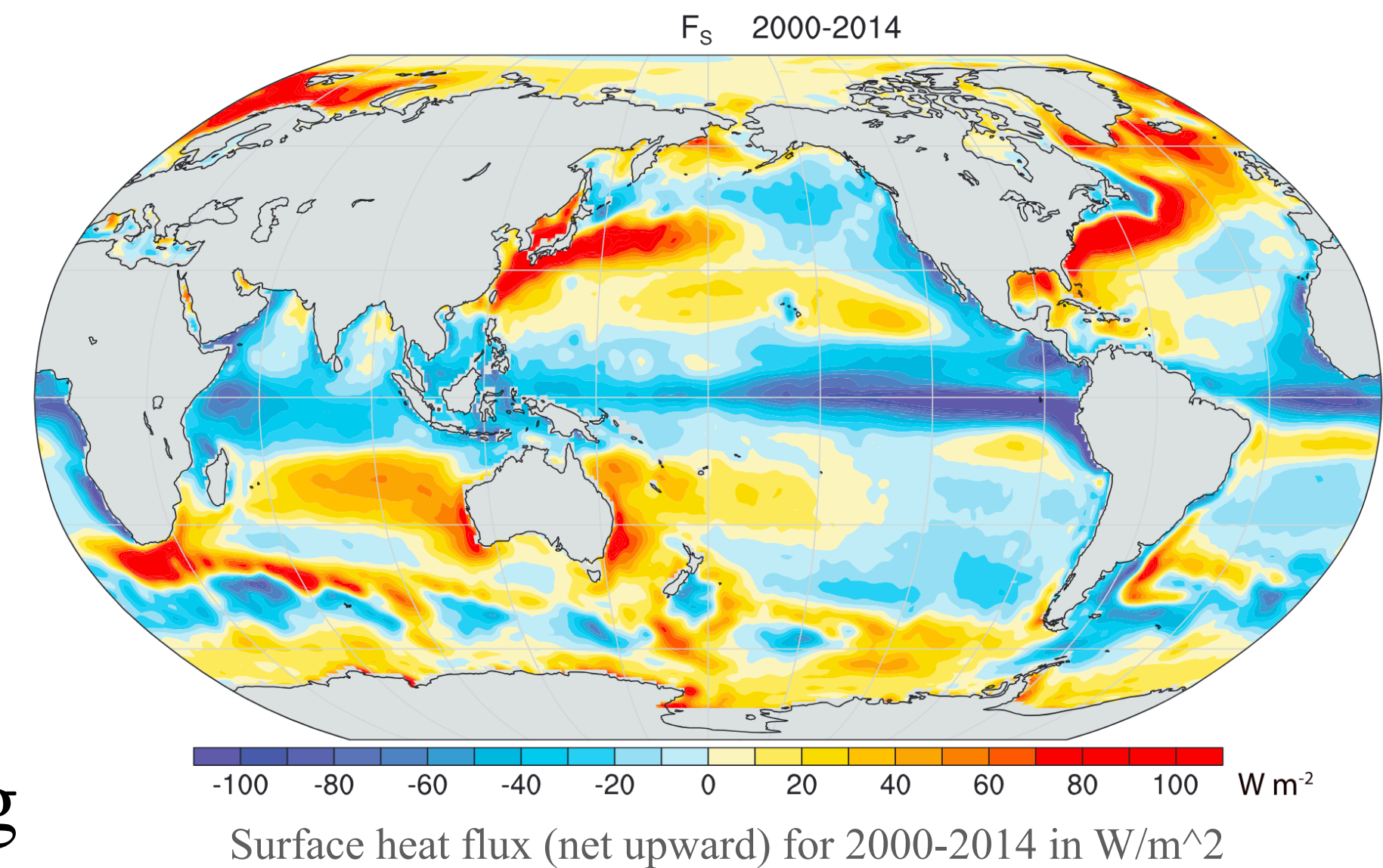
$$F_s = -dOHC/dt - \nabla \cdot \mathbf{F}_O$$

$dOHC/dt$: change in ocean heat content, $>0 \Rightarrow$ ocean warming

∇F_O : heat transport convergence (HTC), $>0 \Rightarrow$ divergence, transport away

$\Rightarrow$ Rearranging for ∇F_O and zonal integration yields:

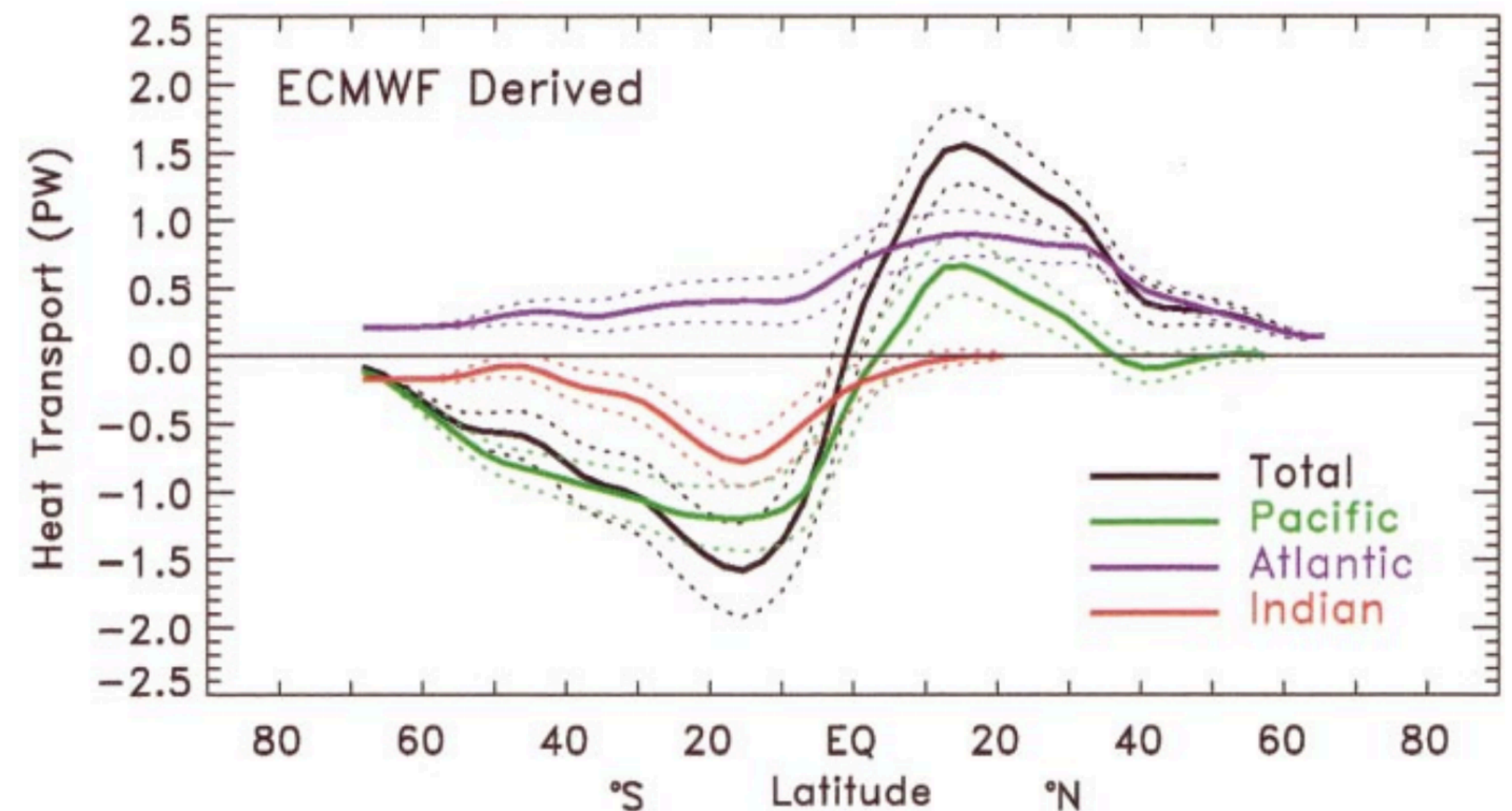
$$MHT(\phi) = \int_{\phi}^{90} [F_s + dOHC/dt] a d\phi$$



Motivation

The Atlantic: A special case

- **Atlantic is unique:** Heat transport is northward at all latitudes
- This transport is dominated by the **AMOC** (Atlantic Meridional Overturning Circulation)



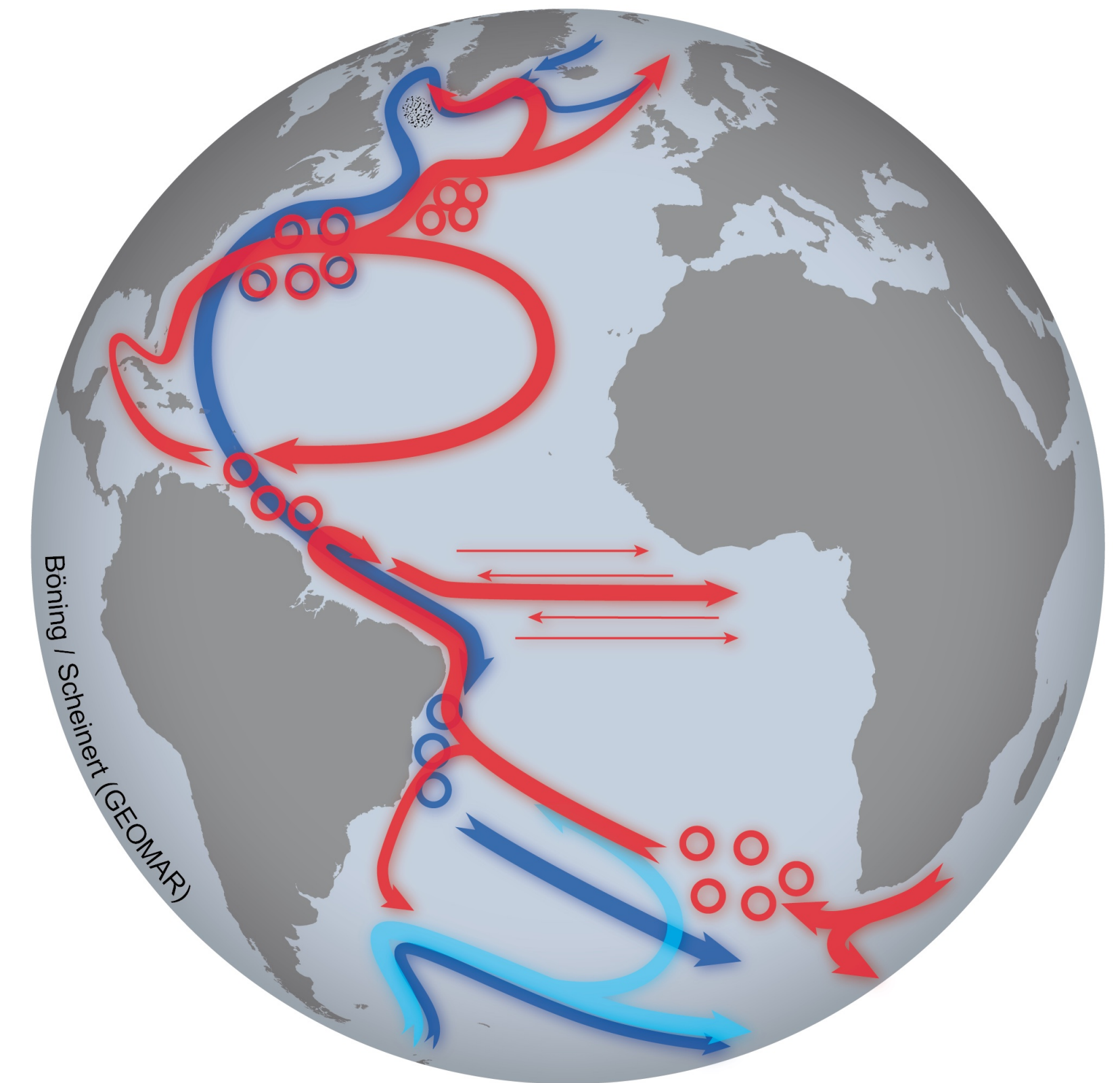
Global estimates for heat transport, *Trenberth and Caron (2001)*

Motivation

What is the AMOC?

AMOC:

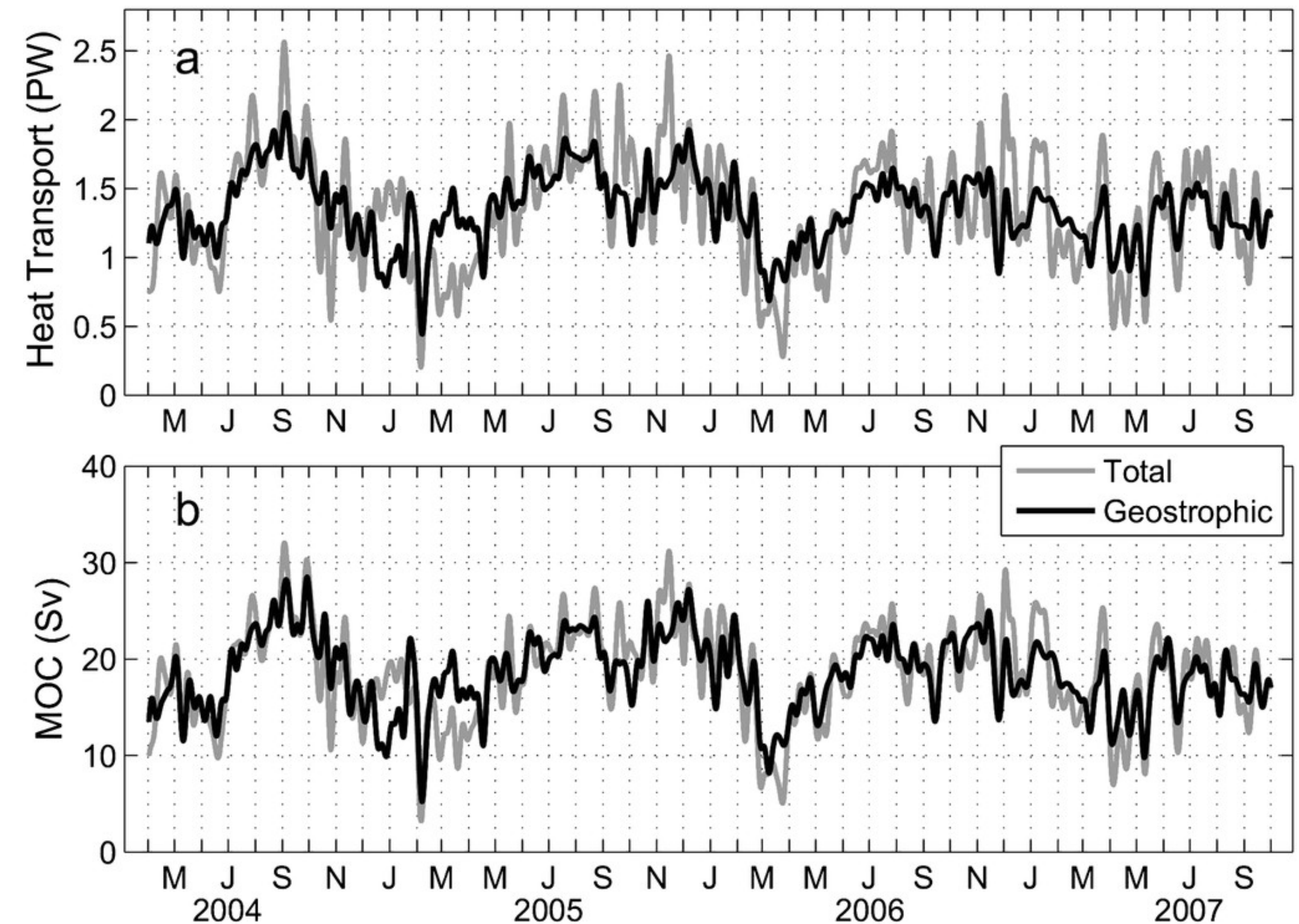
- Warm surface water flows northward
 - Deep convection close to Arctic
 - Transported back south at depth
 - Connects with gyre circulations and eddies
 - Responsible for most heat transport in Northern Hemisphere
- contributes to Europe's relatively mild climate compared to other regions at similar latitudes
- (National Oceanography Center, <https://www.noc.ac.uk>)



Motivation

The Atlantic: A special case

- **Atlantic is unique:** Heat transport is northward at all latitudes
- This transport is dominated by the **AMOC** (Atlantic Meridional Overturning Circulation):
 - *Johns et al. (2011)* found **nearly linear relationship** between **MHT-AMOC at 26°N**
 - And AMOC carries around ~88% of total MHT in Atlantic



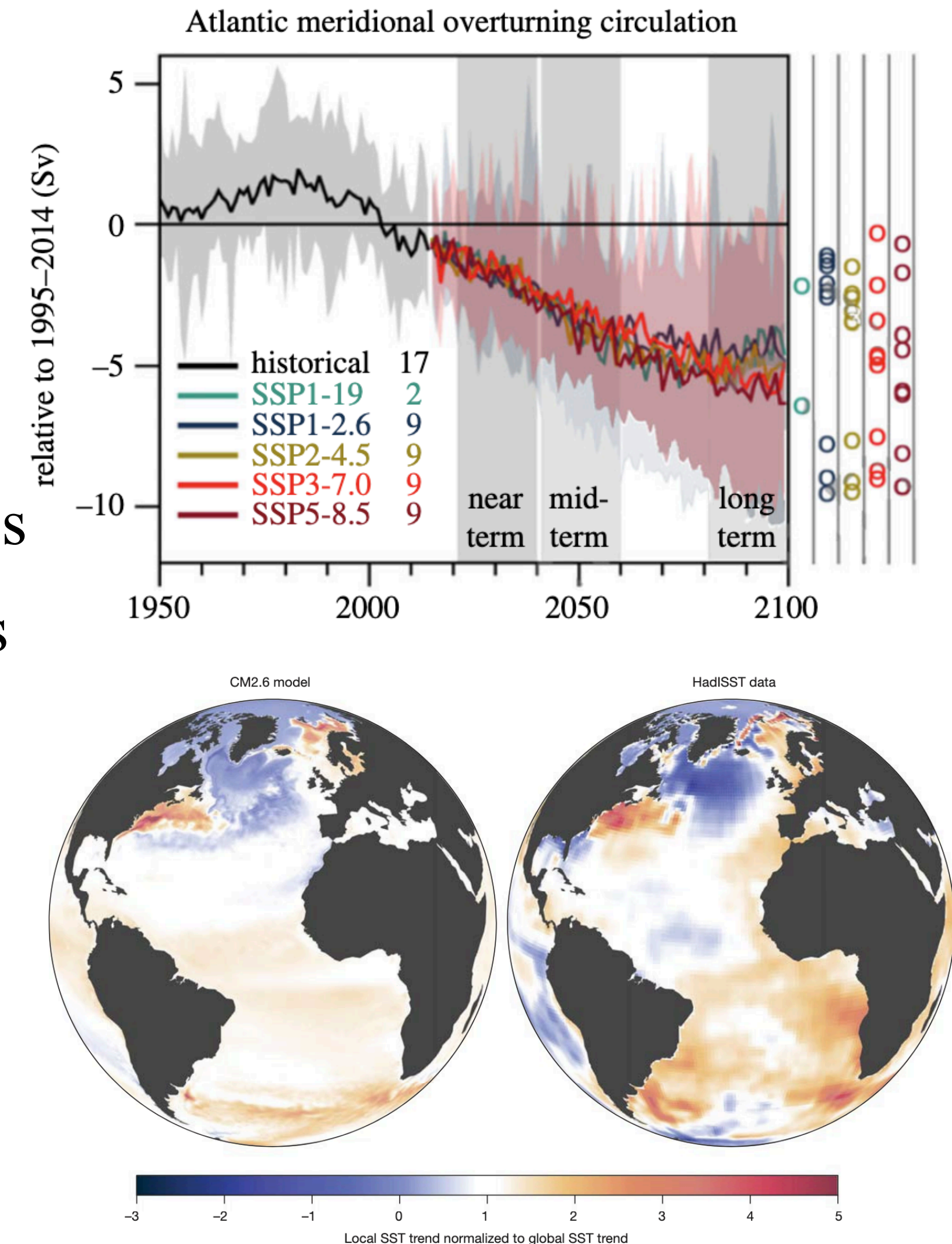
Heat Transport and MOC at 26°N, *Johns et al. (2011)*

Impacts

What could happen for weakening AMOC and MHT?

- Climate models predict AMOC weakening in this century (*IPCC (2021), WG I, Chapter 9*)
→ decrease in MHT would be expected
- Weaker MHT could result in:
 - Less heat transported northwards and accumulation in the tropics
 - Cooling in the North Atlantic (cold blob) → colder temperatures over Europe
 - Change in position of ITCZ, southward shift (*Marshall et al. 2013*) ...

Can we confirm this with observational data?



Research Gap

Current observations are insufficient to distinguish forced trends from natural variability

- Observed 0.17 PW MHT reduction at 26°N for 2009-2016, *Bryden et al (2020)* — but is this a **forced trend or natural variability**?
- Interannual variability driven by **climate modes**
→ e.g. AMOC/MHT drop in 2009, following record low NAO

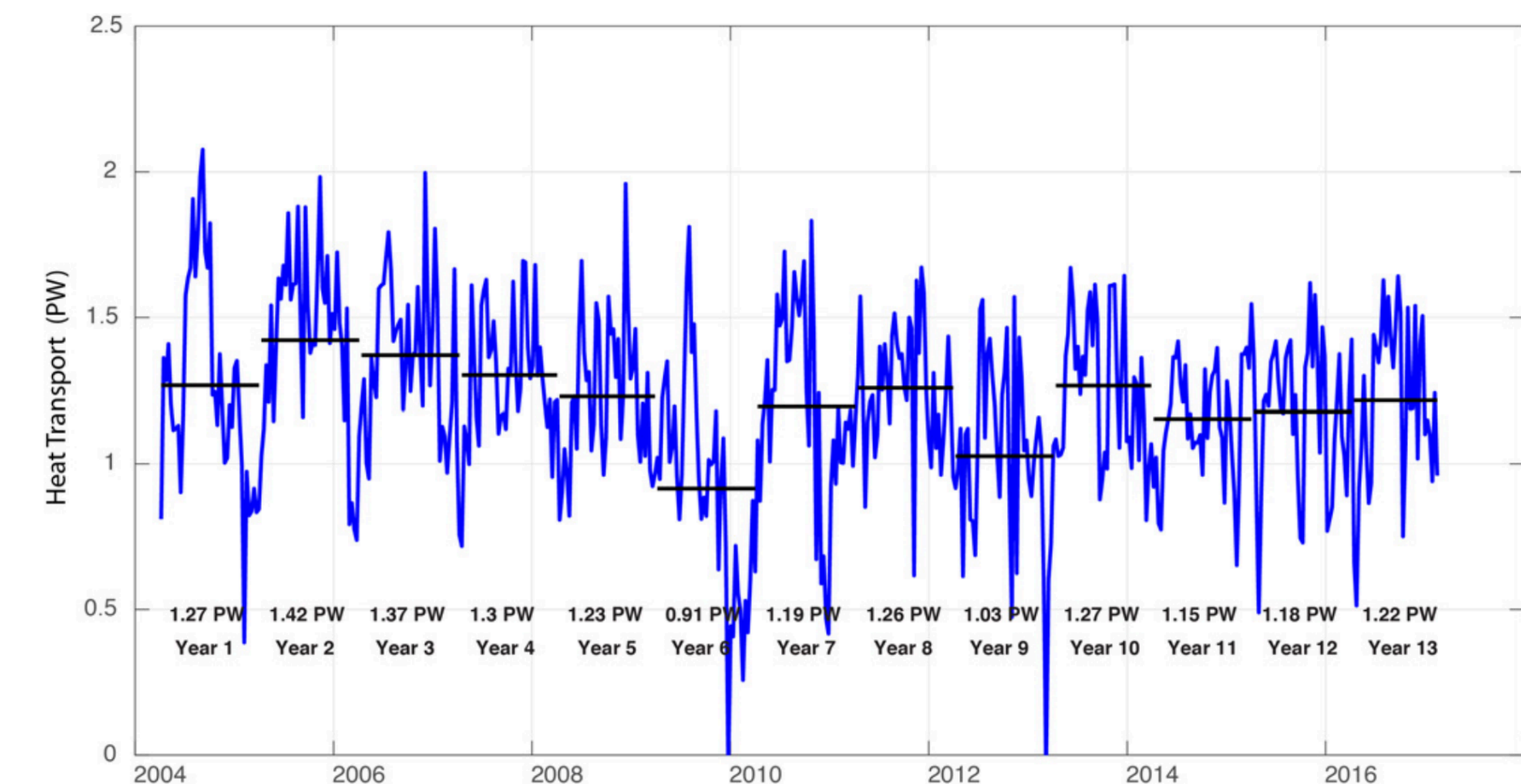


Figure from *Bryden et al (2020)*

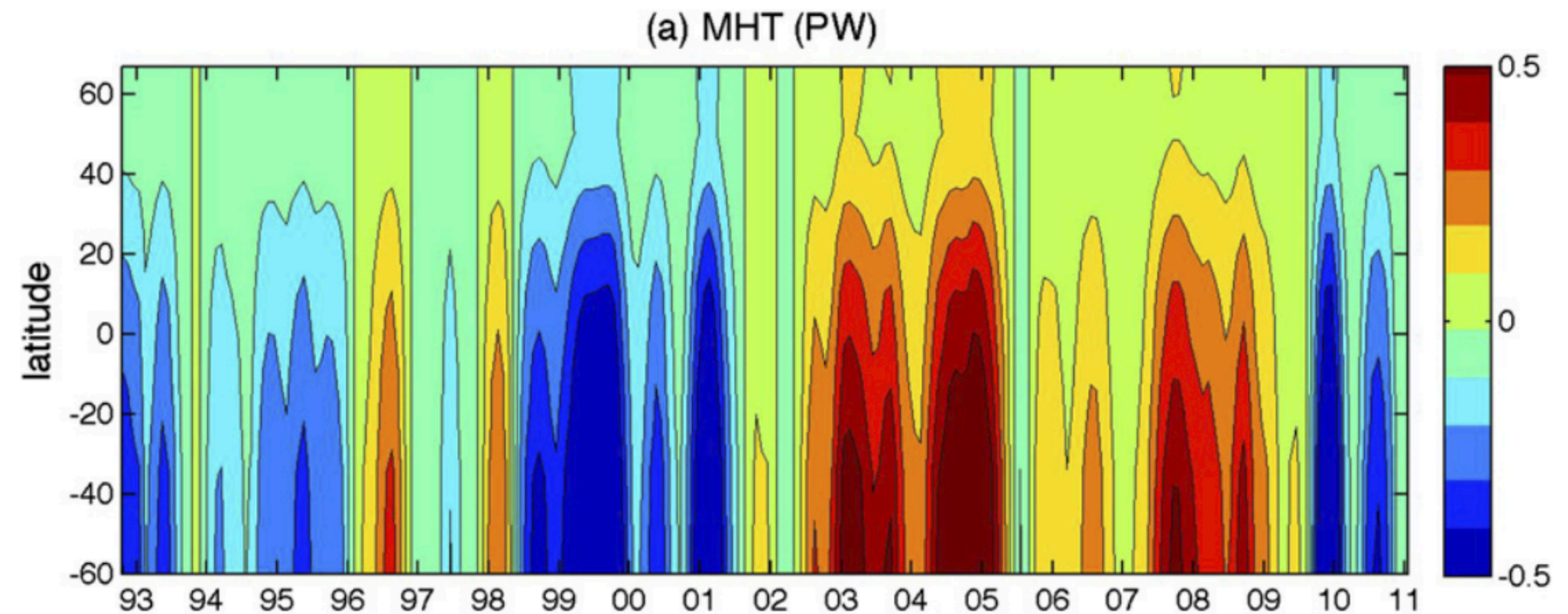
Open Questions:

1. **Observational limitations:** continuous mooring arrays are **short and spatially sparse**
2. AMOC-MHT connection well-established for 26°N → **basin-wide latitudinal MHT coherence poorly understood**
3. To be able to **detect anthropogenic forced trend** we first need to understand its **natural variability**

Thesis objectives

Investigating basin-wide coherence of Atlantic MHT

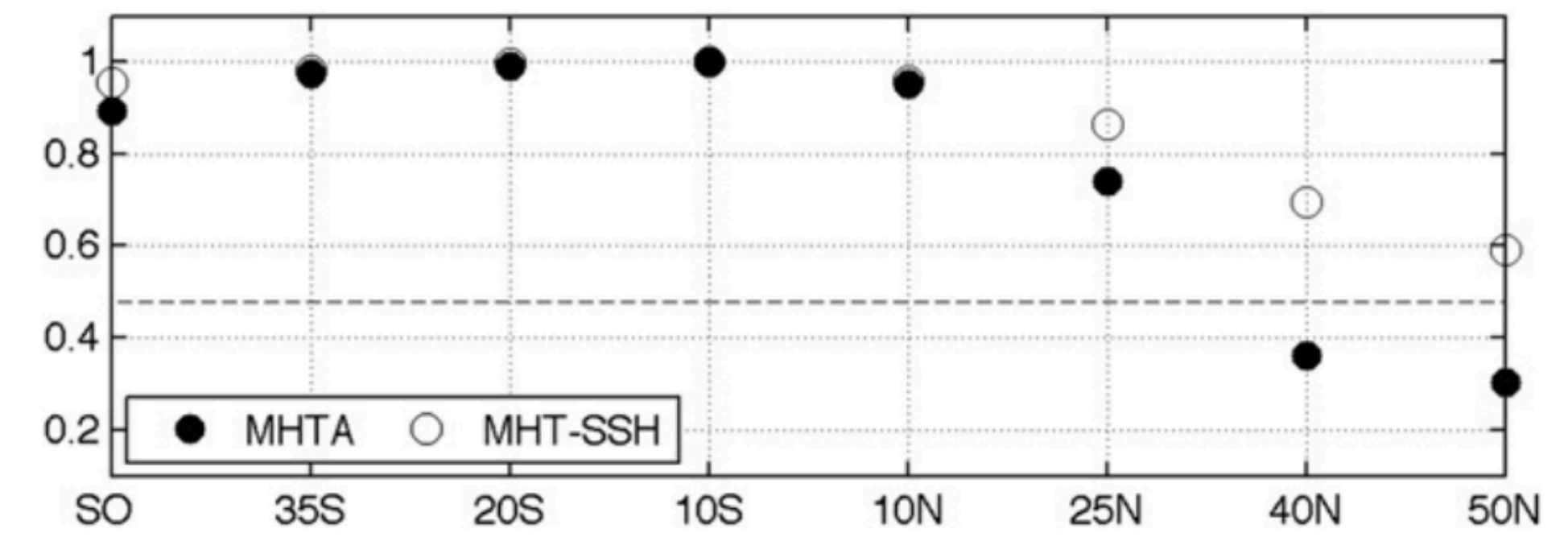
1. What is the spatiotemporal coherence of Atlantic MHT variations?
2. Is incoherence between latitudes caused by changes in circulation or in surface heat flux?



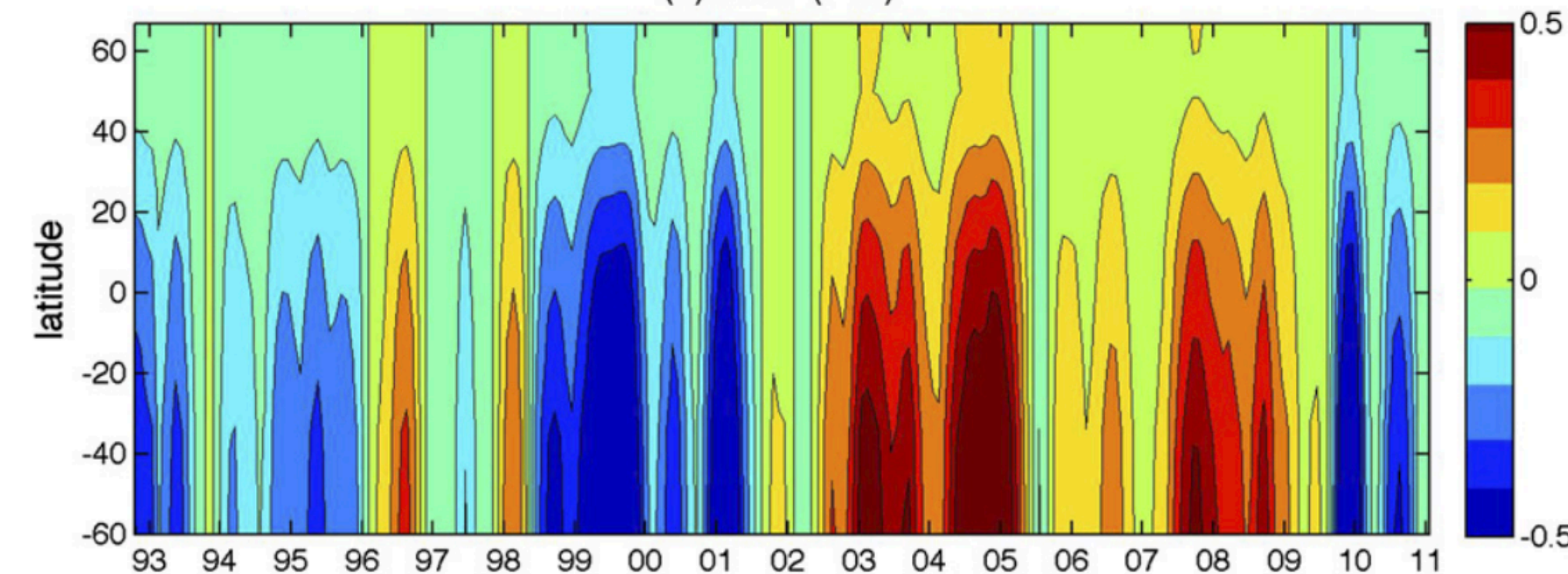
Thesis objectives

Context: What is coherence?

- Coherence: Anomalies covarying across latitude bands
- *Kelly et al. (2014)* found coherence from 35°S until 40°N
 - High correlation at 0 lag between bands
 - Dominant timescale is interannual (lower figure)
- **Open question:** Does the signal truly arise simultaneously or does it propagate through the basin with a lead/lag relationship?
- Record ends 2011 → extend by using *Calafat et al. (2025)* estimates for MHT until 2020



(a) MHT (PW)



Upper: Spatial coherence of MHT, correlation with 10°N, Lower: First mode of variability of MHT anomalies (PW), *Kelly et al. (2014)*.

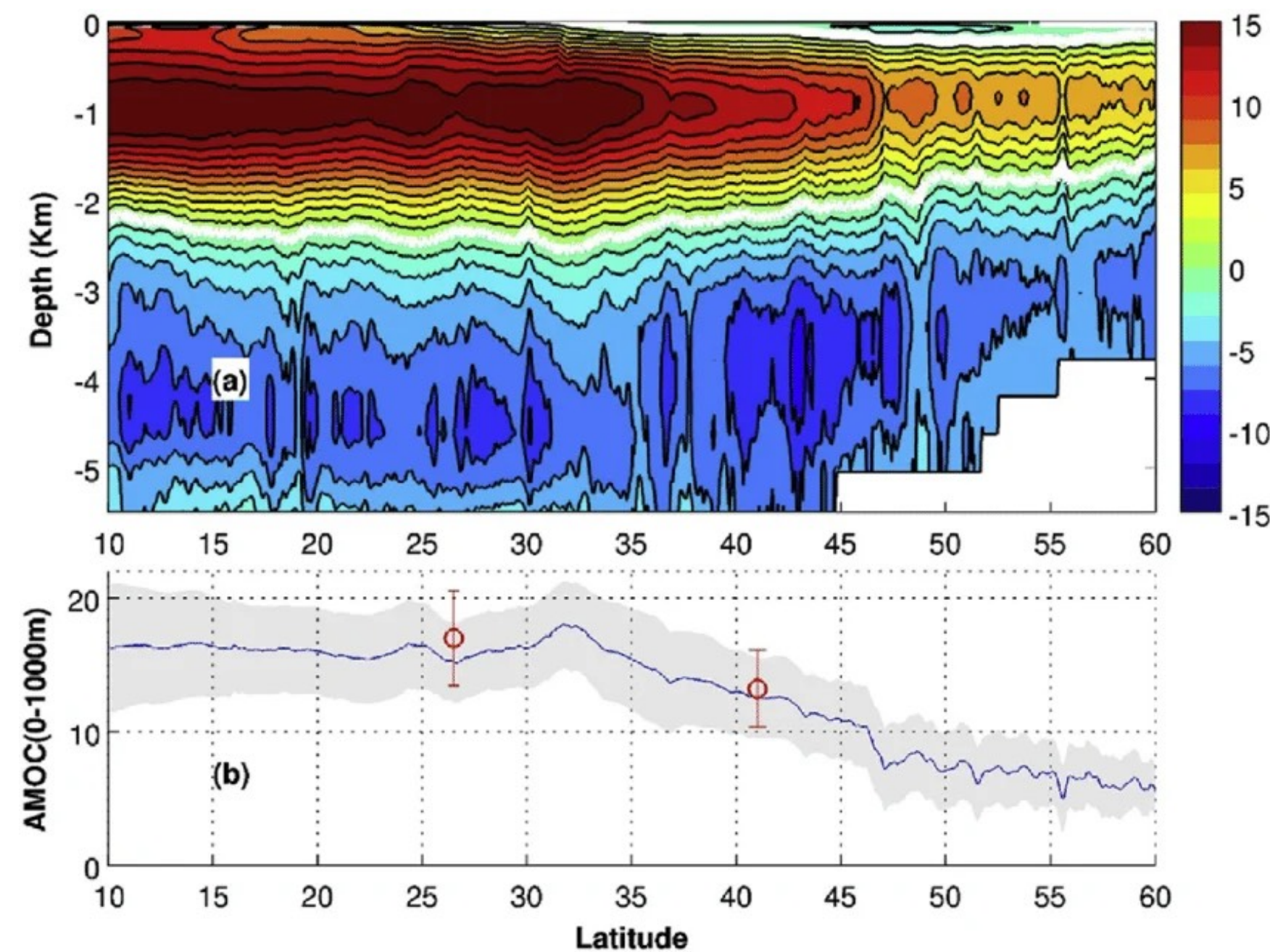
Thesis objectives

Decomposing MHT into its drivers

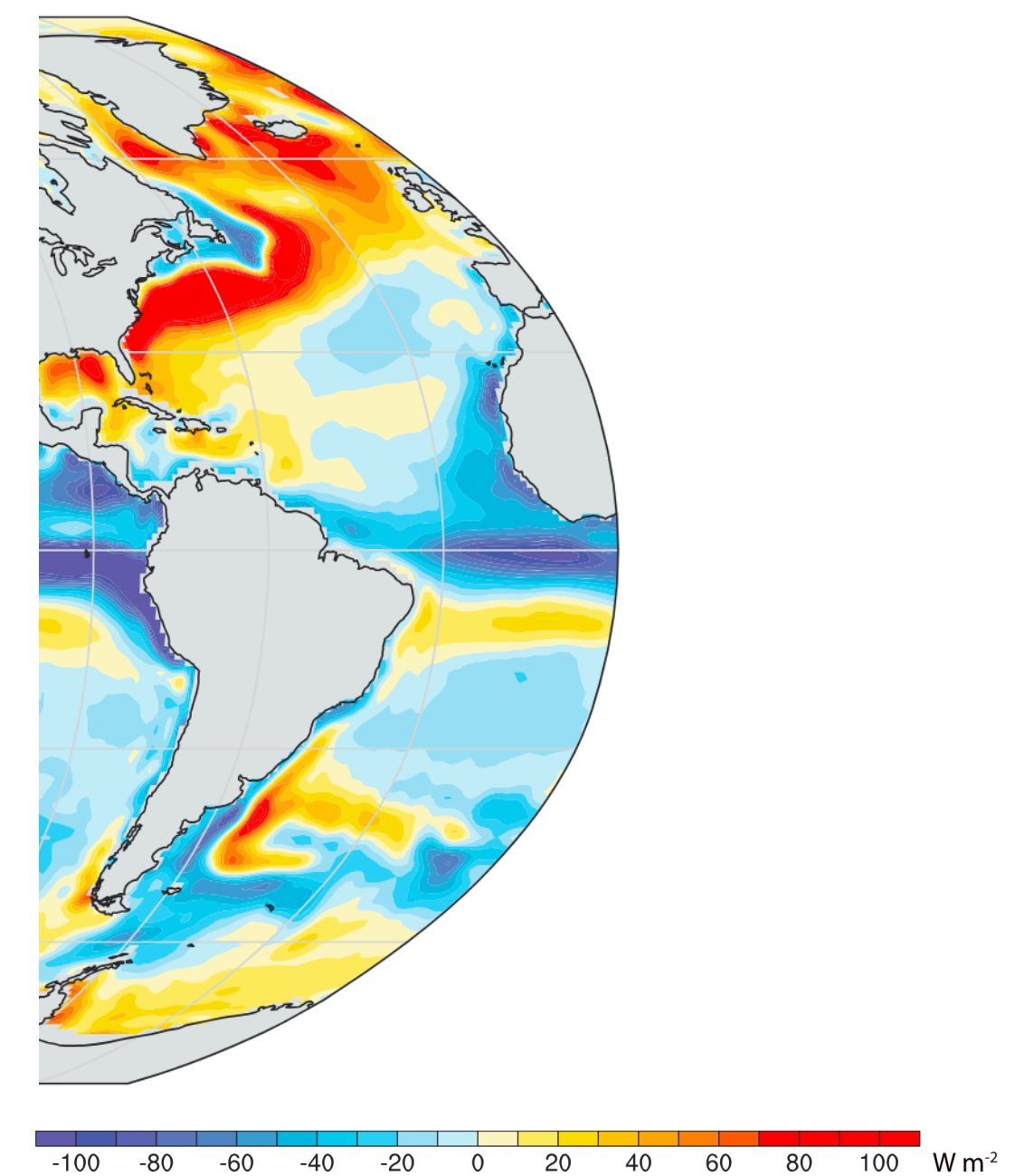
1. What is the spatiotemporal coherence of Atlantic MHT variations?
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Open question:

What drives incoherence between latitude bands and how do they connect to climate indices?



Mean AMOC Streamfunction and mean transport



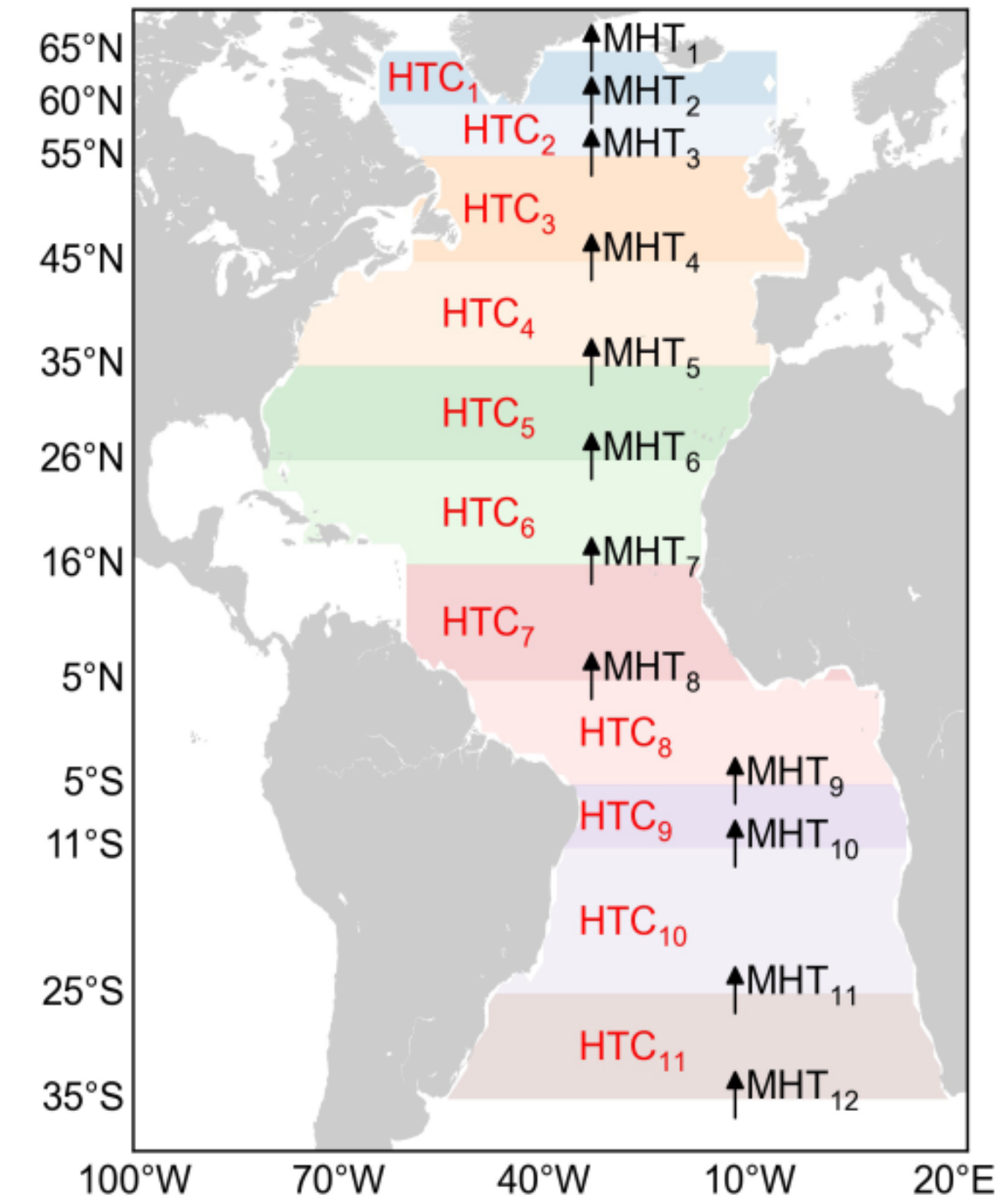
Surface heat flux (net upward) for 2000-2014 in W/m²

Methods - Data

MHT estimates from the Bayesian Hierarchical Model (BHM)

MHT estimates from Calafat et al. (2025):

- Combines **hydrographic data** (argo) with **satellite data** (altimetry and gravimetry)
 - Cost-efficient, high latitudinal resolution
- Different datasets are combined by using **Bayesian Hierarchical Model**
 - Accounts for different spatial resolutions between datasets and **uncertainty propagation**
- Dataset offers:
 - Probabilistic estimates for MHT and Heat Transport Convergence (HTC) for
 - **12 latitude bands** from **35°S to 65°N**
 - From 2004-2020 at 3-month resolution



12 Latitude bands, Calafat et al. (2025)

Methods - Data

How is MHT calculated using the BHM?

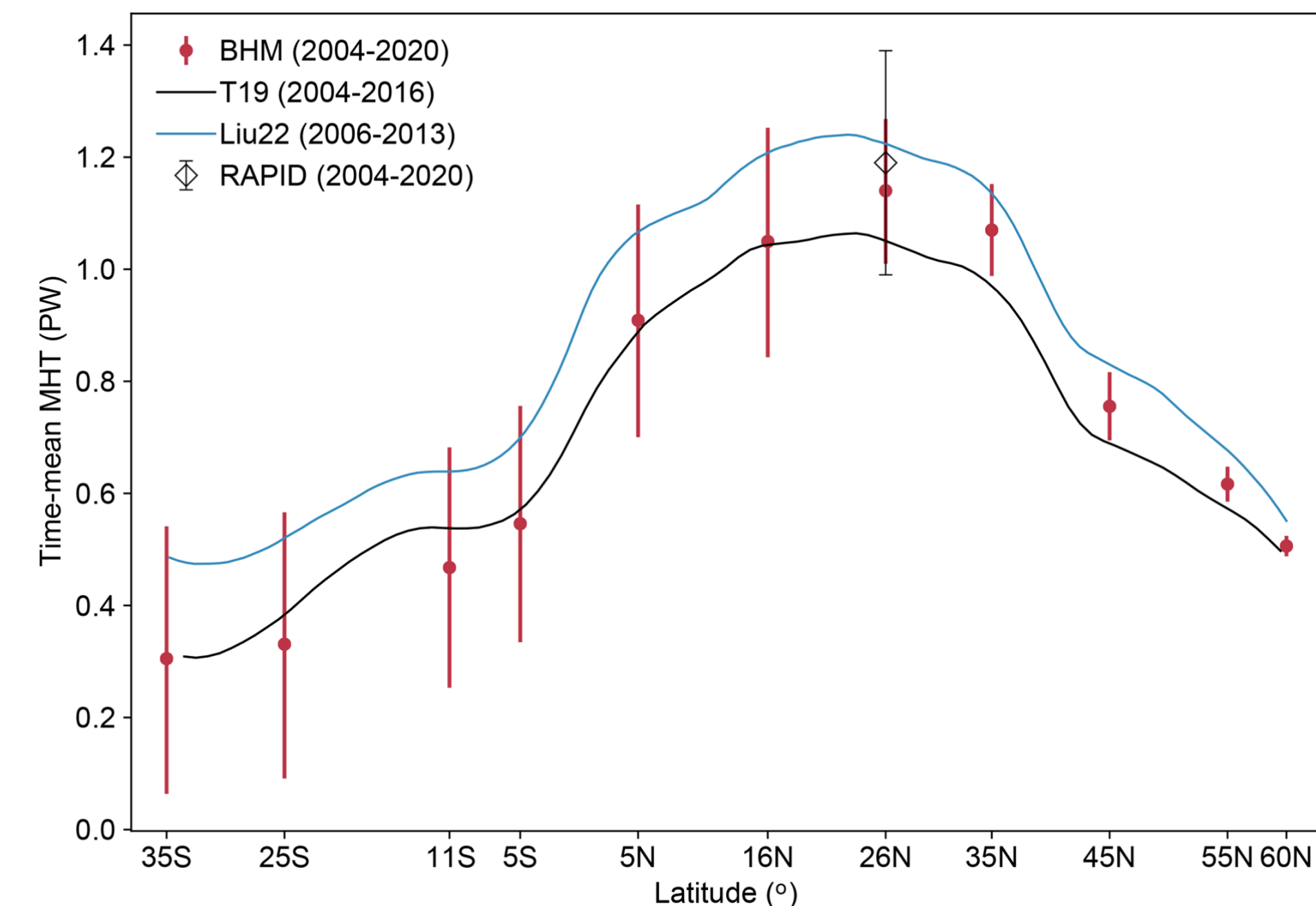
- MHT indirectly inferred by meridionally integrating HTC

$$\text{MHT}_{i,t} = \text{MHT}_{1,t} + \sum_{j=1}^{i-1} \text{HTC}_{j,t}, \quad i = 1, \dots, M + 1,$$

- HTC is computed as the residual of Ocean Heat Content (OHC) tendency - surface heat flux (F_S)

Comparing to other datasets:

- BHM estimates capture MHT well (e.g. at 26°N)
- Comparable to other methods
 - Trenberth et al. (2019)* for 2004-2016 and *Liu et al. (2022)* for 2006-2013
 - Both: Observation based using earth's energy budget



First look at the data

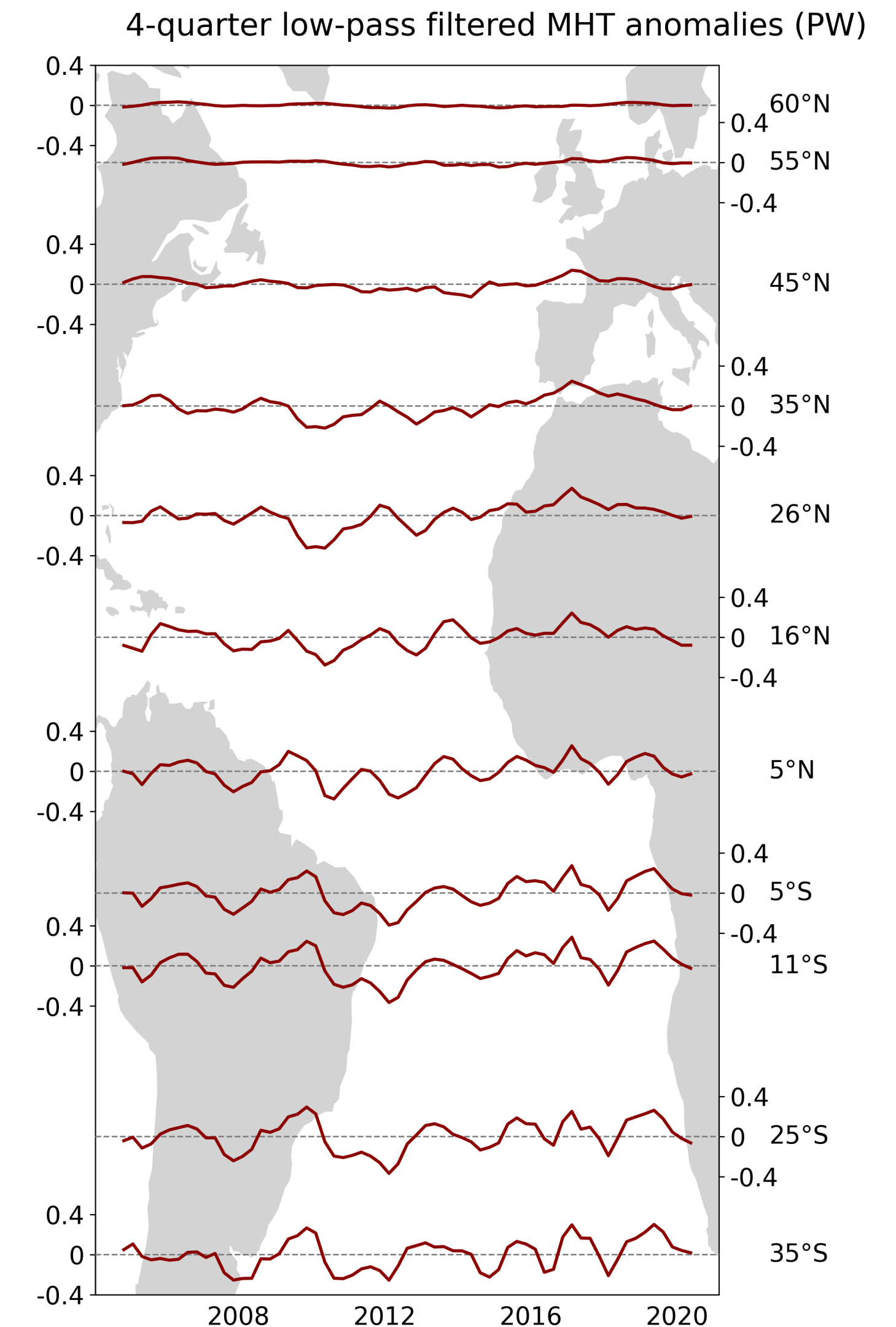
Processing and initial observations

Processing the data:

- Deseasonalized
- 4-quarter low-pass filtered (running mean)

First glance impressions:

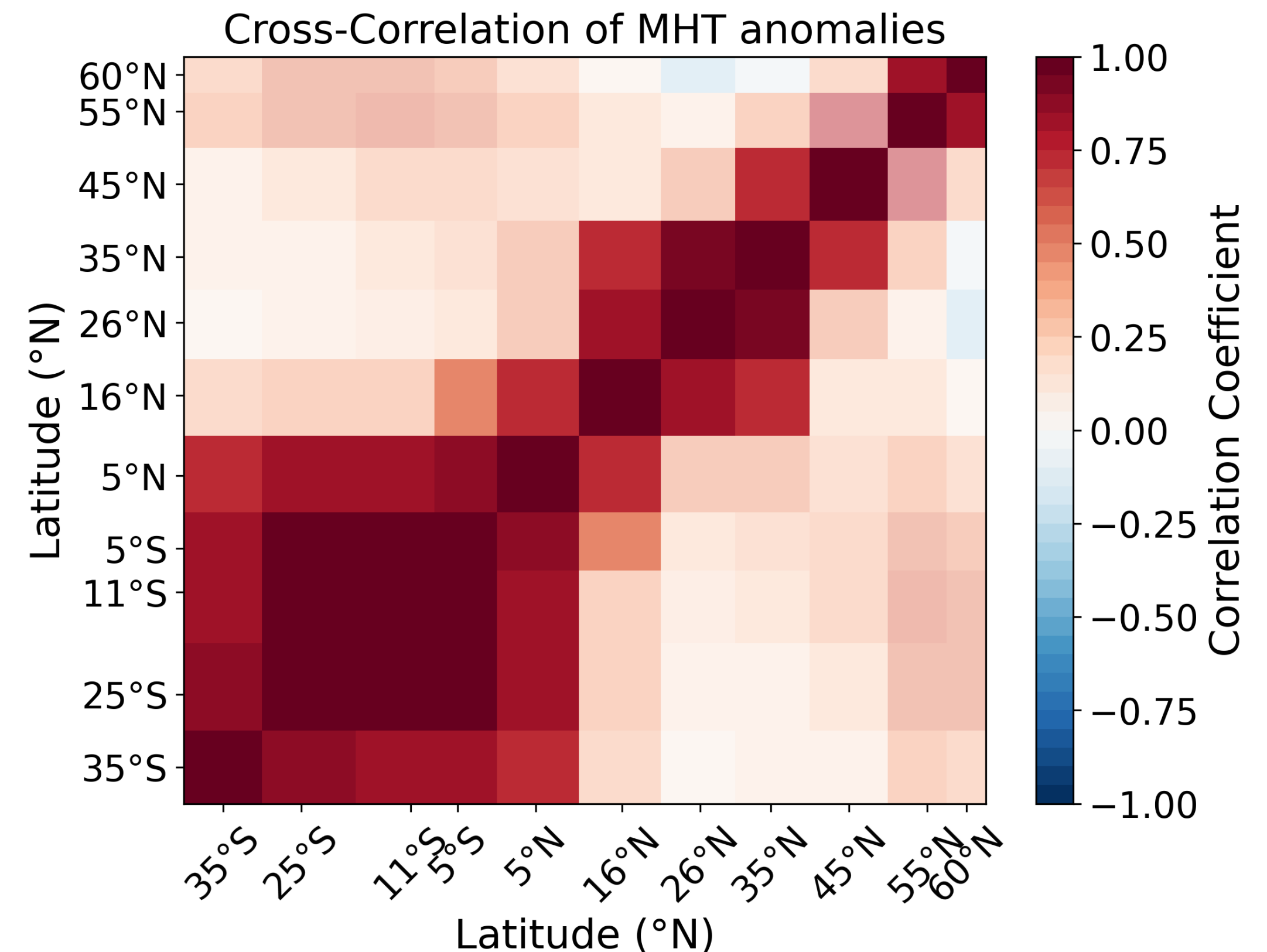
- Smallest anomalies at northern latitudes — likely reflecting the MHT=0 constraint at 65°N
- Further south the variability and the uncertainty increase



Reproducing Calafat et al. (2025)

Covarying latitude bands in agreement

- Calafat found two covarying bands: 35 - 16°N and 5°N - 35°S
- Cross correlation plot between all latitude bands reveals:
 - Two covarying bands clearly visible
 - Correlation is significant
- High-latitude North Atlantic not part of either band
 - Consistent with *Kelly et al. (2014)* coherence breakdown north of 40°N



Analysis approach and next steps

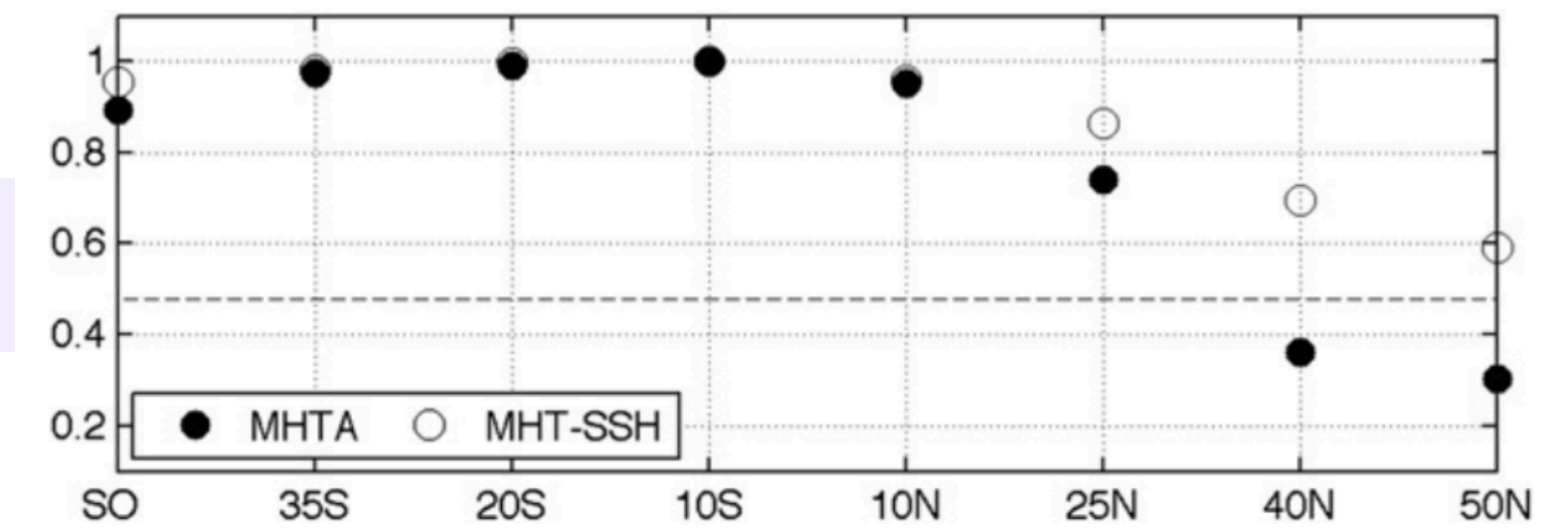
Investigating basin-wide coherence of Atlantic MHT

1. What is the spatiotemporal coherence of Atlantic MHT variations?

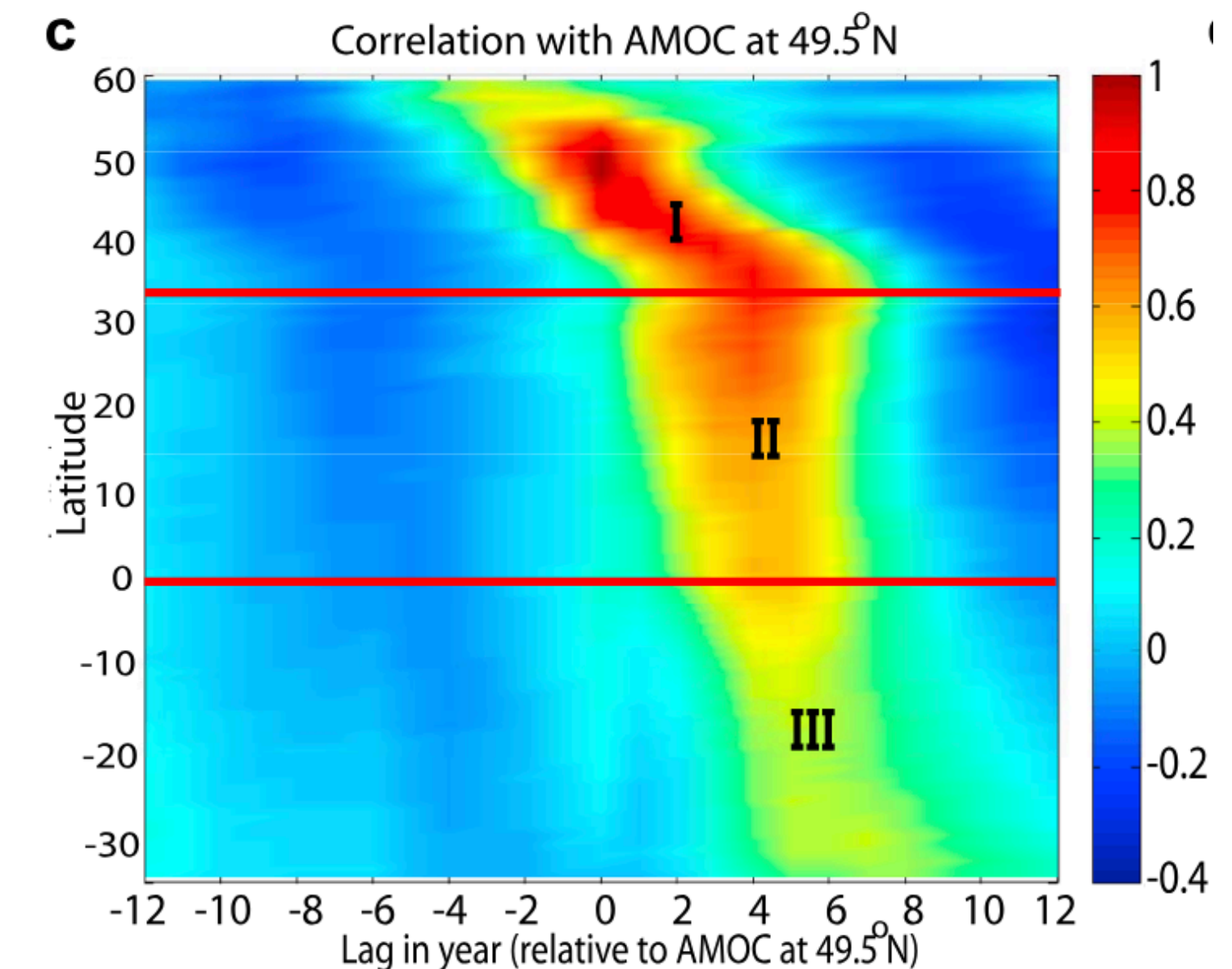
- 0-lag analysis (*Kelly et al. (2014)*)
- Lead/lag analysis: Signal propagation between latitude bands (*Zhang et al. (2010)* approach)
- Including significance testing

2. Is incoherence between latitudes caused by changes in circulation or in surface heat flux?

- Decompose MHT variability into circulation and surface heat flux contributions
- Correlate to climate indices (e.g. NAO)



Spatial coherence of MHT, correlation with 10°N, *Kelly et al. (2014)*.



Correlation between AMOC variations at 49.5°N and AMOC variations at all latitudes over the entire control simulation, *Zhang et al. (2010)*.

Summary

Investigating basin-wide coherence of Atlantic MHT

Motivation/Research Gap:

- Ocean stores 91% of excess energy from EEI
- Observed MHT reduction at 26°N — forced or natural variability?
- MHT basin-wide latitudinal coherence/signal propagation poorly understood

Objectives:

- What is the spatiotemporal coherence of Atlantic MHT variations?
- What drives incoherence?

Data:

- Probabilistic MHT estimates from Calafat et al. (2025)
- 12 Latitude bands for the time period 2004-2020

